From Pi to Pythagoras' Theorem to Hippocrates' Lune

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Abstract

This derivation of Pythagoras' Theorem illustrates the fundamental relationship between the sides of a right triangle and inscribed circles. The construction of the proof in this paper implements the circle's area over that of the unit square.

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1 Introduction

The construction of a proof of Pythagoras' Theorem begins with a unit right-angle triangle. A circle centered on A with radius $r = 1$ is drawn, and similarly, a circle centered on B with radius $r = 1$ is constructed. Another larger circle is centered on B with radius equal to the *hypotenuse* $\overline{BC}$.

The constant π is defined as the ratio of a circle's circumference C to its diameter d:

$$\pi = \frac{C}{d}.$$

Consequently, the area of a circle with radius r is

$$A_{\text{circle}} = \pi r^2.$$

Let us observe that the area of the circle centered at B with radius $\overline{BC}$ is $A_{\text{circle},BC}$. Removing the area of the smaller circle centered at B with radius $\overline{BA}$ leaves the area of the annulus:

$$A_{\text{annulus}} = A_{\text{circle},BC} - A_{\text{circle},BA}.$$

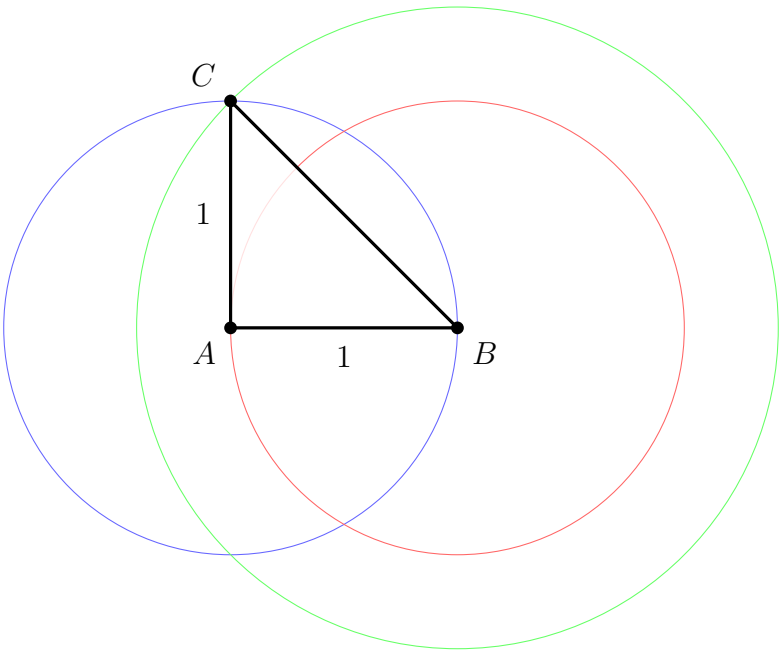


Figure 1: A right triangle with unit base and height, together with circles centered at the base vertices.

The area of the annulus is the focus of interest until it is observed that $\overline{AC}$, tangent to the smaller circle centered at B, simultaneously and concurrently sweeps the radius $\overline{AC}$ of the circle centered at A. The area of the annulus is therefore equal to the area of the circle centered at A with radius $\overline{AC}$.

$\overline{AC}$ acts as the relative tangential sweep of the annulus as well as the absolute radial sweep of the circle in the area of interest.

$$A_{\text{annulus}} = A_{\text{circle},BC} - A_{\text{circle},BA}$$
$$\pi AC^2 = \pi BC^2 - \pi BA^2.$$

Rearrange the terms and divide both sides by π to obtain the Pythagorean relationship for an equilateral right-angle triangle:

$$AB^2 + AC^2 = BC^2.$$

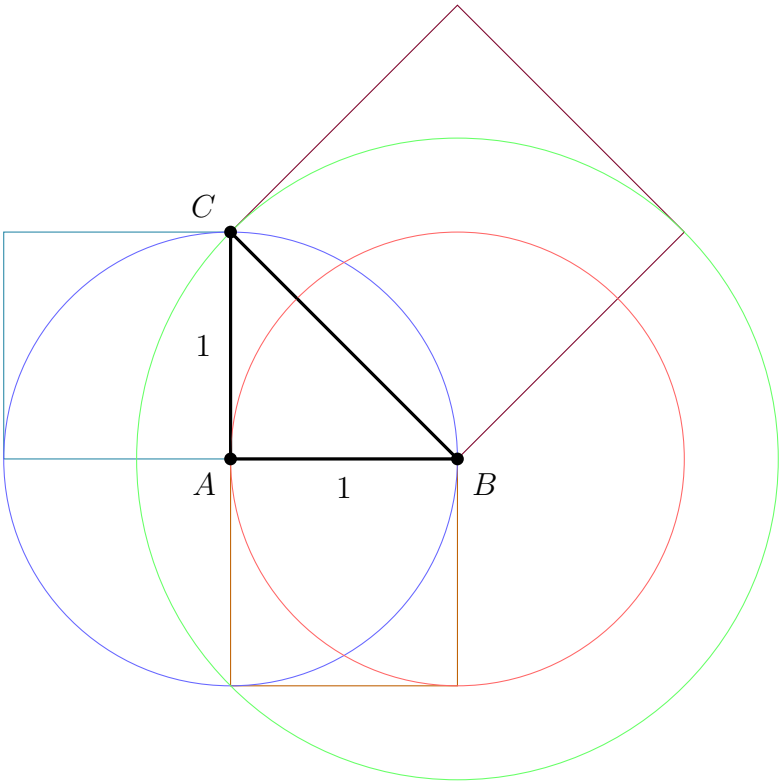


Figure 2: Additional squares added to the sides of the right-angle triangle.

The Pythagorean relationship exists as a consequence of the area of circles where one side of the right angle triangle that is also the radius of the circle centered at A coincides with the sweep of the annulus that is the difference of the two circles centered at B whose radii are the base and hypotenuse.

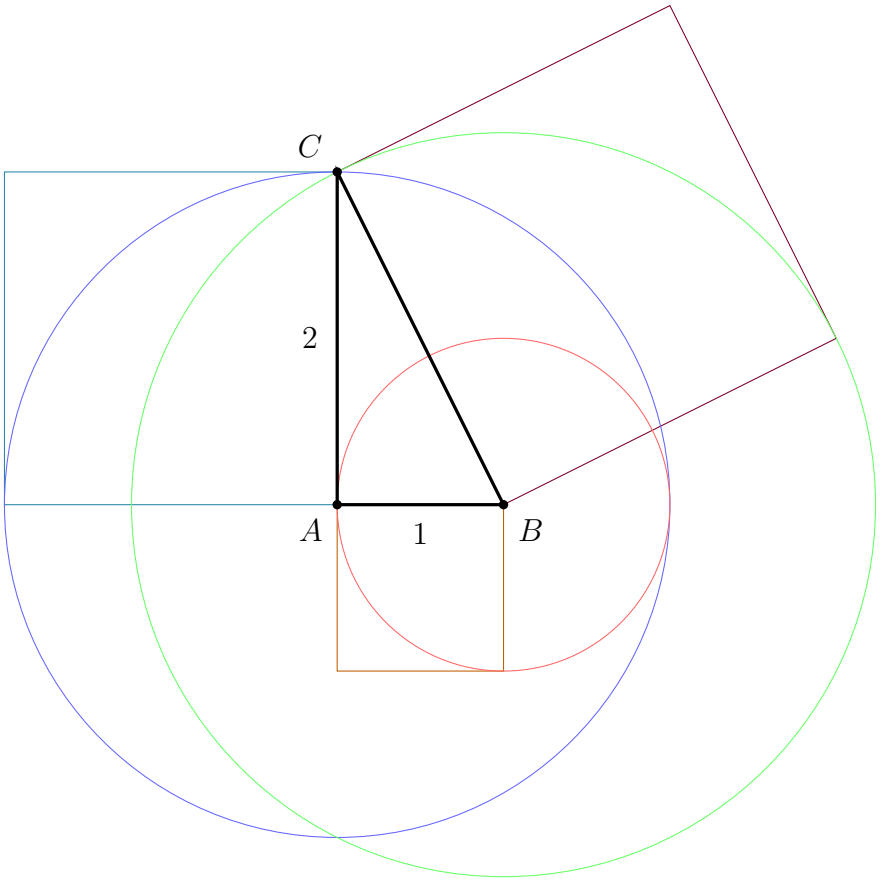


Figure 3: Right-angle triangle with double unit height simultaneously sweeps circle A and annulus.

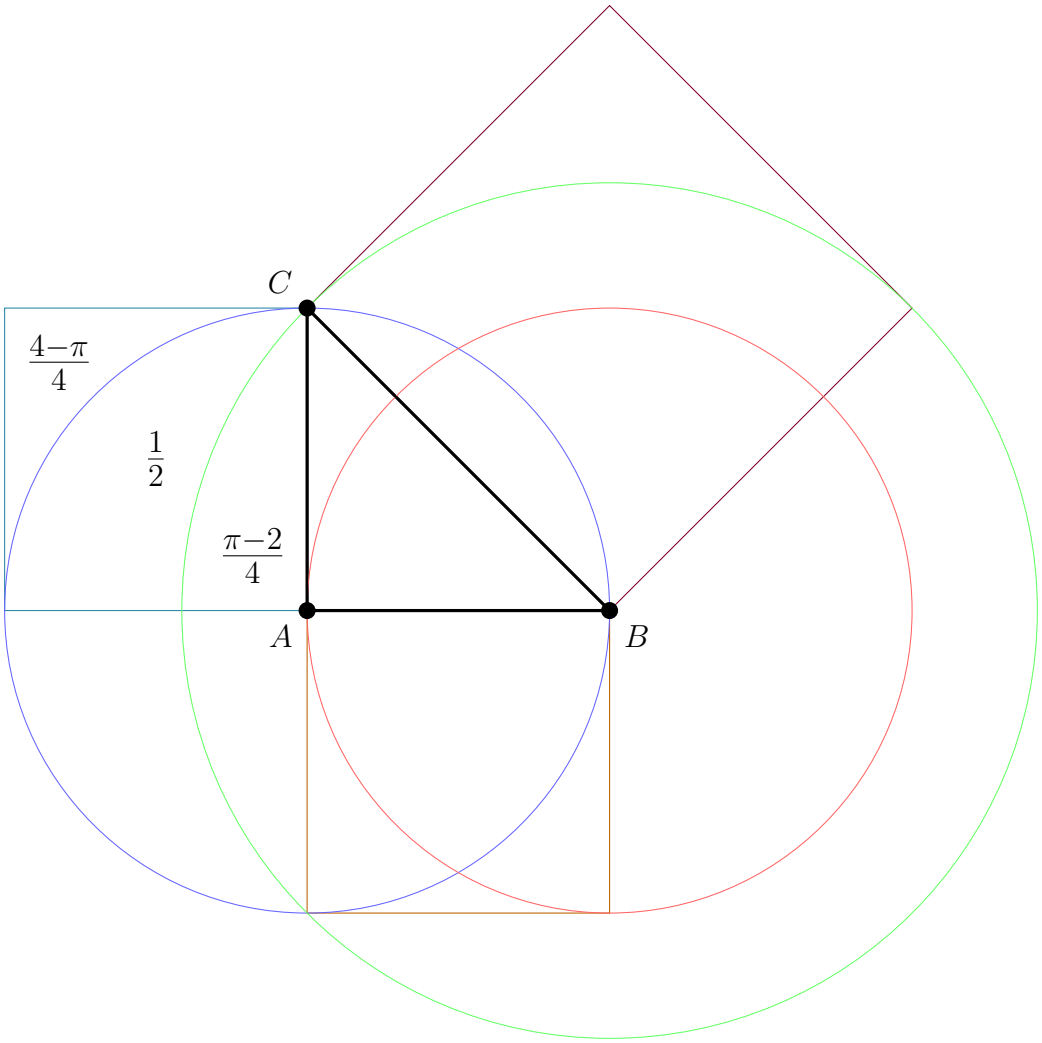


Figure 4: Triangle BAC with $|\overline{AB}| = |\overline{AC}| = 1$. The areas inside the square on $\overline{AC}$ are identified relative to circle A and the annulus.

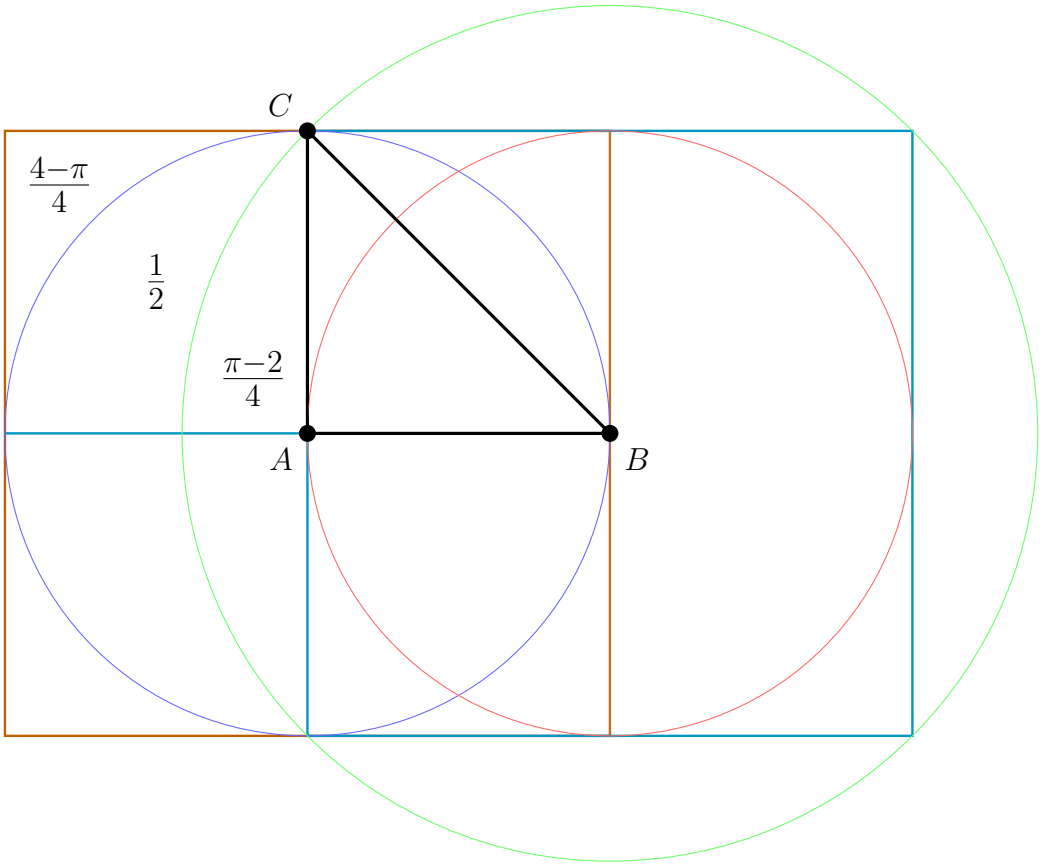


Figure 5: Additional square centered on B.

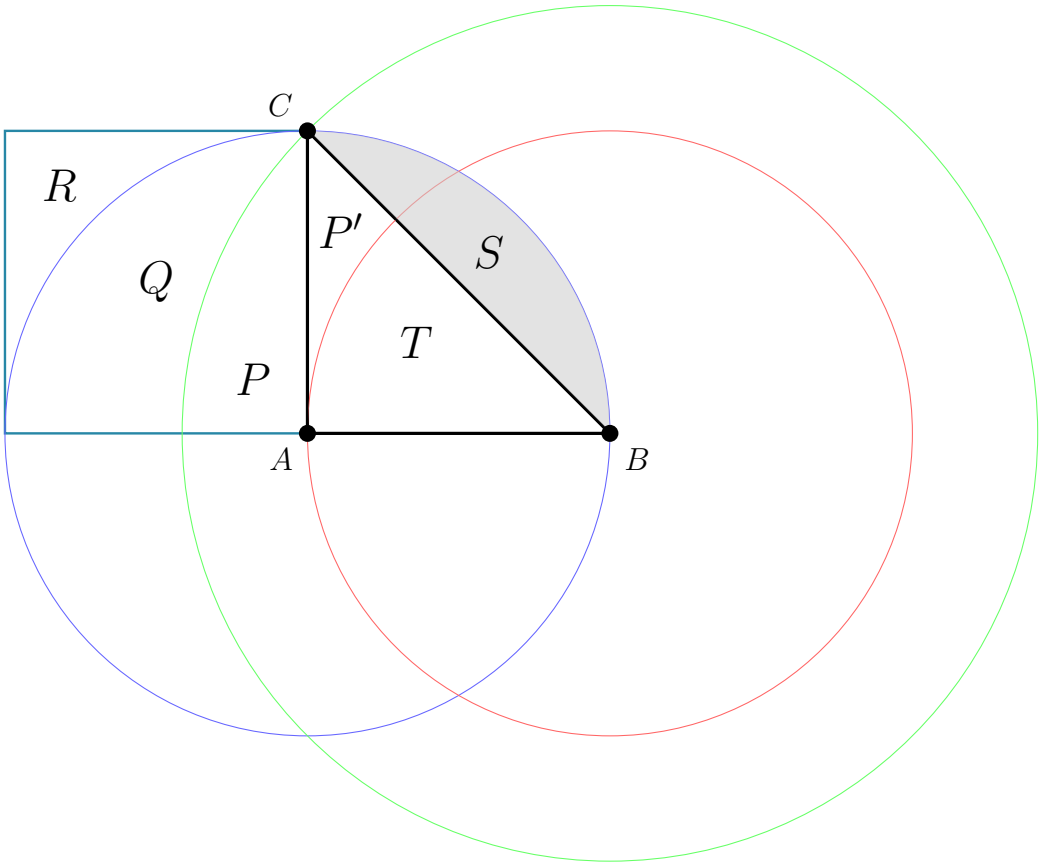


Figure 6: All areas identified from P to T. S identifies the shaded area.

The classical construction of Hippocrates' lunes is discussed by Girstmair 2003 and Bogomolny 2018.

To find areas of other regions follow the same principle of finding the differences between areas. Finding the area within the quadrant squares of AB and AC that is outside circle centered on A, is determined to be the area of the square A less the area of circle A.

$$A_P = \frac{A_{\text{annulus}} - A_{\text{square},B}}{8} = \frac{\pi - 2}{4}$$

$$A_T = \frac{\pi}{8}$$

$$A_{P'} = \frac{1}{2} - A_T = \frac{\pi}{8} - A_P = \frac{4 - \pi}{8}$$

$$A_R = \frac{4 - \pi}{4}$$

$$A_S = \frac{\pi - 2}{4}$$

$$A_P = A_S$$

$$A_Q = 1 - A_P - A_R = 1 - \frac{\pi - 2}{4} - \frac{4 - \pi}{4} = \frac{1}{2}$$

$$A_Q = A_{triangle,BAC} = \frac{1}{2}$$

$$A_{P'} = \frac{1}{2} A_R$$

2 Further Research

A brief search returned two references that are worth exploring further.

References

Bogomolny, Alexander. *Hippocrates' Squaring of Lunes*. 2018. URL: https://www.cut-the-knot.org/pythagoras/FaultyLuneQuadrature.shtml.

Girstmair, Kurt (2003). "Hippocrates' lunes and transcendence". In: *Expositiones Mathematicae* 21.2, pp. 179–183. DOI: 10.1016/S0723-0869(03)80018-X. URL: https://doi.org/10.1016/S0723-0869(03)80018-X.